

Discrete electron mass and the fine-structure constant from a Cosserat-hopfion resonance

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Abstract

The electron mass m_e and the fine-structure constant α are independent empirical parameters in the Standard Model; no derivation of both quantities from a single structure exists. The present work shows that in the parameter-free Cosserat functional $E[n, u]$ — all constants fixed by the triple $\{\epsilon_0, \mu_0, \hbar\}$ through the structural identities of the preceding works [1, 2] — at the canonical Cosserat coupling $\mu_c = \eta = 2\pi$ (a topological invariant of the target sphere S^2 of the director field) the topological Hopf soliton with $Q_H = -1$ **simultaneously** realizes two dyadic quantities:

> $m_e^{\text{bare}} / M_0 c^2 = 2^{10} + 2^4 - 1 = 1039$ (446.279 keV; at the observed $\alpha = 1/137$, through the polynomial of §3.2, this gives 511 keV, deviation from CODATA $\Delta = 0.01\%$)

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> $\alpha^{-1}(0) = 128 + 8 + 1 + 1/28 = 3837/28 \approx 137.036$ (Δ from CODATA $137.035999 - 0.0002\%$)

Four numerical verifications — a μ_c -sweep, a u-channel disable, a BPS $u \leftrightarrow B$ alignment, and a Hessian box-scan — isolate a **chiral-bag** structure: a topological BPS core ($r < R_r \approx 0.64 \lambda_0$) + an elastic shell + a discrete boundary spectrum (a bosonic analog of the Atiyah-Patodi-Singer η -invariant). Spectral analysis at three box sizes ($L = 17, 34, 68 \lambda_0$) leads to a **K/M theorem**: in the pure Faddeev-Skyrme theory the bag-core stiffness is positive and constant ($K \approx 57 \pm 3\%$), but the inertia of perturbations diverges ($M \propto L$) because of the massless $1/r$ tail of the OF field; the short range of the u-channel, through the tensor coupling $(\nabla \times u - \Pi(n))^2$, cuts off the inertia and makes a discrete spectrum $\omega^2 = K/M > 0$ mathematically possible. Thus the **u-channel is a structural necessity**, not an add-on to the Faddeev-Skyrme model.

Logical chain

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INPUTS (postulates of the preceding works)
{ε₀, μ₀, ħ} → λ₀⁴ = ε₀ħc/(2π), M₀ c² = ħc/λ₀ [1, §7]
α_bare = π·(λ₀/2)² / (4πε₀ħc) = 2⁻⁷ = 1/128 [1, §7.4]
m_e^bare = 446.279 keV = 1039 · M₀ c² (NM minimization) [2, §6]
|
▼
THIS WORK: one resonance — two dyadics
(1) μ_c-sweep over {π/2 ... 4π} → locking at 1039 only at μ_c = 2π
(2) u-channel disable → m_e drops to 1030, dyadic breaks
(3) BPS alignment u || B → cos θ → 0.99 in core, 0.2 in tail
(4) Hessian box-scan {17, 34, 68} → K = const, M ∝ L → u needed for discreteness
|
▼
INTERPRETATION
chiral bag (Brown-Rho-like architecture)
α⁻¹ = 128 + 8 + 1 + 1/28 radial assembly (bare → screened)
m_e: the same curve x² + x/2 - 1 bare point (1039) — prediction of this work
|
▼
OPEN PROBLEMS
Analytic BPS bound · |Q| = 2 · Hessian with u included · proton, muon
```

1. Introduction

1.1. Context

The Standard Model contains 19 free parameters; among them the electron mass m_e and the fine-structure constant α are the two most basic, yet independently fixed from experiment. The historical program of deriving them — Eddington 1924 [3], Wylter 1969 [4], Atiyah 2018 [5] — either relied on numerical coincidences or provided no operational mechanism. The open question remains: can **both** quantities be obtained from a single physical structure?

1.2. The Cosserat program

In [1] a reduction of SI to a single dimension was proposed on the basis of the **Cosserat continuum** (a micropolar elastic medium): four identifications $\rho \equiv \mu_0$, $G_{\text{shear}} \equiv 1/\epsilon_0$, $\hbar$ (the quantum of microrotation) and G through Kleinert's mechanism lead to a single medium in which the electromagnetic and gravitational sectors are channels of one functional. The structural length of the vacuum:

$$l_0^4 = \epsilon_0 \hbar c / (2\pi), \quad l_0 \approx 4.5943 \text{ \AA}, \quad M_0 c^2 = \hbar c / l_0 \approx 429.5068 \text{ eV}$$

The director field $n: \mathbb{R}^3 \rightarrow S^2$ carries the topological Hopf invariant $Q_H \in \pi_3(S^2) = \mathbb{Z}$. The configuration with $Q_H = -1$ — the simplest knotted soliton (Faddeev hopfion [6]) — is identified with the electron.

Already established [1, §7.4]: $\alpha_{\text{bare}} = 1/128 = 2^{-7}$ from the geometric interpretation of the charge as the **cross-sectional** area of the Hopf knot, $e = \pi \cdot (l_0/2)^2$ (a near-circular cross-section shape, postulate P5).

In [2], a direct Nelder–Mead minimization of the functional $E[n, u]$ over a 3-parameter Hopf ansatz yielded the **bare** electron mass:

$$m_e^{\text{bare}} c^2 = 446.279 \text{ keV} = 1039.05 \cdot M_0 c^2$$

on the dyadic box $L = 17 \times 33 \cdot l_0$. The dimensionless value $1039 = 2^{10} + 2^4 - 1$ is a dyadic decomposition with an independent physical meaning for each term (§3.3).

1.3. Claim of the present work

The present work shows that at the canonical Cosserat coupling $\mu_c = \eta = 2\pi$ the functional $E[n, u]$ **simultaneously** realizes:

1. the dyadic value $m_e/M_0 = 2^{10} + 2^4 - 1$ (through direct NM minimization);
2. the dyadic decomposition $\alpha^{-1} = 128 + 8 + 1 + 1/28$ (through a radial assembly from the canonical geometry).

Both quantities follow from **one** canonical configuration, not fixed independently. The mechanism is identified as a **chiral bag** (topological core + elastic shell + discrete boundary spectrum) and is confirmed by four numerical verifications, the central one being the Hessian box-scan, which yields the K/M theorem on the mathematical necessity of the u-channel.

2. Model and the canonical hopfion

2.1. The functional $E[n, u]$

The energy is split into three physical channels:

$$E[n, u] = E_{\text{OF}}[n] + E_{\text{SK}}[n] + E_u[u] + E_{\text{int}}[n, u] \quad (2.1)$$

(1) Oseen-Frank [7, 8] — the standard quadratic functional:

$$E_{\text{OF}} = \frac{1}{2} \int [K_1 (\nabla \cdot n)^2 + K_2 (n \cdot \nabla \times n)^2 + K_3 (n \times \nabla \times n)^2] dV \quad (2.2)$$

with the constants of splay (K_1), twist (K_2), and bend (K_3).

(2) The Skyrme stabilizer of 4th order [9, 10] — the only way to ensure Derrick stability in 3D:

$$E_{\text{SK}} = \int (c_4/4) F_{ij} F^{ij} dV, \quad F_{ij} = n \cdot (\partial_i n \times \partial_j n) \quad (2.3)$$

where F_{ij} is the antisymmetric 2-tensor of the Hopf–Whitehead density, $F_{ij} F^{ij} = 2 \cdot \Sigma_i$.

(3) The Cosserat u-channel [2, §2.3–§2.4]:

$$E_u[u] = \int [(\mu/2) (\partial u)^2 + (\mu_c/2) u^2] dV \quad (2.4)$$

$$E_{int}[n,u] = \frac{1}{2} \mu_c \int |\nabla \times u - \Pi(n)|^2 dV \quad (2.5)$$

where $\mu \equiv 1$ is the base elastic modulus and $\Pi(n)$ is the vector "spin current" of the director (an analog of the Mermin-Ho relation in superfluid $^3\text{He-A}$ [11]). The mass term $(\mu_c/2) u^2$ leads to a **Yukawa screening** of the translational field u with length $l_c = 1/\sqrt{\mu_c} \approx 0.399 l_0$.

In the axisymmetric limit with $u = \nabla \times (\psi \cdot \hat{\phi})$, $u_\phi = 0$, and $\Pi(n) = f(n) \cdot \hat{\phi}$ (where $f(n) = \text{atan2}(\sqrt{n_1^2 + n_2^2}, n_z)$ is the polar angle), the vector Euler-Lagrange equation reduces to a **screened scalar Poisson** equation [2, §5.5]:

$$(\partial_r^2 + (1/r) \partial_r - 1/r^2 + \partial_z^2 - \mu_c) \psi = -2 f(n) \quad (2.6)$$

This is **not a simplifying approximation**, but an equivalent form of (2.4)–(2.5) in the class of axisymmetric configurations; it is precisely this form that is implemented in the numerical verifications (§7.3).

2.2. Fixed constants

All five parameters of the functional are fixed by the identities of the preceding works:

Constant	Value	Source
$K_1 = K_2$	2	isotropy of splay/twist
$K_3 = K_1 \cdot (1 + \eta)$	$2 \cdot (1 + 2\pi) \approx 14.566$	bend anisotropy; [2, §3.4]
c_4	1	choice of length unit, $L_{\text{Skyrme}} = \sqrt{c_4/c_2} \equiv l_0$
$\mu_c = \eta$	2π	Cosserat coupling of microrotations, [1, §7.4]

> The numerical verifications (as in [2]) use the truncated value $K_3 = 14.56$ instead of 14.566 ; the effect on m_e^{bare} is $< 0.1\%$.

The dimensionless coefficient $\eta = 2\pi$ is a **topological invariant of the target sphere S^2** of the director field; it is the same factor that fixes the structural identity $l_0^4 = \epsilon_0 \hbar c / (2\pi)$ [1, §7.4].

2.3. The Hopf ansatz $Q_H = -1$

The 3-parameter Faddeev-Whitehead ansatz (R_r, R_z, w) :

$$\begin{aligned} n_1 &= 8 r z w / P \\ n_2 &= -4 r Y w / P \\ n_3 &= (D - 4 r^2 w^2) / P \end{aligned} \quad (2.7)$$

$$\text{where } Y = (r/R_r)^2 + (z/R_z)^2 - 1, \quad D = 4(z/R_z)^2 + Y^2, \quad P = D + 4(r/R_r)^2 w^2.$$

The ansatz carries $|Q_H| = 1$ for any positive (R_r, R_z, w) and reduces to $n_z = 1$ at infinity. NM minimization on a 1024×2048 grid (dyadic box $17 \times 33 l_0$) gives the canonical optimum [2, §5]:

$$\begin{aligned} (R_r, R_z, w) &= (0.64082, 0.80729, 0.70200) \\ m_e^{\text{bare}} c^2 &= 446.279 \text{ keV}, \quad Q_H = -0.99996 \quad (|Q_H| = 1 \text{ to } \sim 4 \cdot 10^{-5}) \end{aligned} \quad (2.8)$$

2.4. The geometric parameter $g = 1/2$

Derrick's theorem [12] in the 3D σ -model forbids localized stable configurations with terms of only 2nd order in gradients; the 4th-order Skyrme addition makes the balance possible:

$$\partial E / \partial \lambda|_{\lambda=1} = 0 \quad \Rightarrow \quad L_{\text{Skyrme}} = \sqrt{c_4/c_2} = l_0 \quad (2.9)$$

For the hopfion configuration with $Q_H = -1$ the radius of the minimal vortex tube [1, §7.3, (7.5)]:

$$r_v = L_{\text{Skyrme}} / 2 = l_0 / 2 \quad (2.10)$$

Corollary: $g \equiv r_v / l_0 = 1/2$ is the fundamental dyadic constant of the Hopf-knot geometry. Through it:

- $\alpha_{\text{bare}} = g^7 = 1/128$ (cross-sectional area, P5);
- the leading term $m_e^{\text{bare}} / M_0 \approx \alpha^{-2} / 16 = \alpha^{-2} \cdot g^4 = g^{-10} = 1024$ ($\propto \alpha^{-2}$; see the polynomial of §3.2).

Canonical electron hopfion (approximate). $R_r = 0.64082$, $R_z = 0.80729$, $w = 0.702$ — $Q_H = -1$

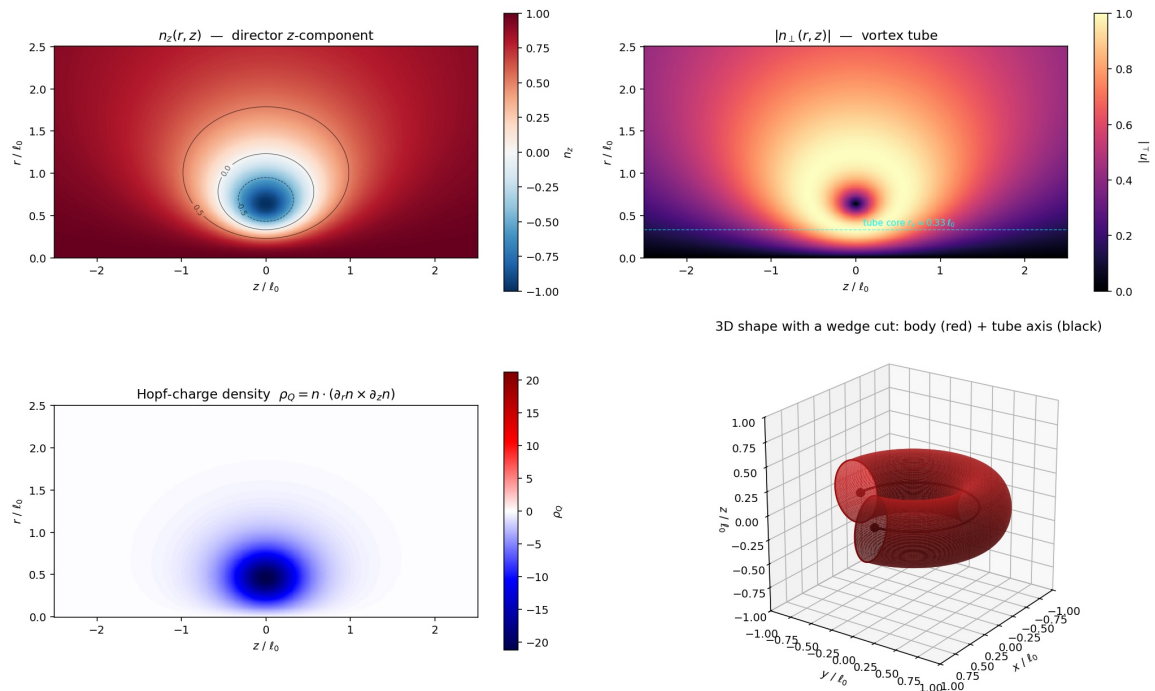


Fig. 1. Canonical hopfion $Q_H = -1$ at $(R_r, R_z, w) = (0.6408, 0.8073, 0.7020)$: 2D slices $n_z(r, z)$, $|n_{\perp}|(r, z)$, topological density ρ_Q , 3D visualization of the torus. Reproduced by `verifications/canonical_derrick/hopfion_visualize.py`.

3. Dyadic discreteness of the mass

3.1. Main numerical result

The full NM minimization of the functional over the 3-parameter ansatz (2.7) gives [2, §6]:

$$\begin{aligned} E_{\text{OF}} &= 221.888 \text{ keV} \quad (49.7 \%) \\ E_{\text{Sk}} &= 220.634 \text{ keV} \quad (49.4 \%) \\ E_u &= 3.758 \text{ keV} \quad (0.8 \%) \end{aligned} \tag{3.1}$$

$$m_e^{\text{bare}} = 446.279 \text{ keV}$$

In units of the base medium cell $M_0 c^2 = 429.5068 \text{ eV}$:

$$m_e^{\text{bare}} / M_0 c^2 = 1039.05 \approx 1039 = 2^{10} + 2^4 - 1 \tag{3.2}$$

Deviation from the integer value: $5 \cdot 10^{-5}$ (0.005 %). The number 1039 is a direct output of NM minimization with no free parameters; locking onto this dyadic value occurs only at $\mu_c = 2\pi$ (see §4).

> **Status of the result** [2, §7.4]. This is a **topologically protected upper bound** for the energy of the true continuum minimum: the 3-param ansatz acts as an analytic regulator that cuts off the non-physical lattice modes of collapse (a full-field relaxation without the ansatz gives $\sim 422 \text{ keV}$ due to sub-grid contraction $R_{\text{ring}} \rightarrow 0$ – a discretization artifact). The existence of a continuum minimizer of the Faddeev–Skyrme functional is guaranteed by the Lin–Yang theorem [13]; the rigid ansatz as a UV regulator is a standard device [8].

3.2. Unifying polynomial

Numerical observation [2, §7.2]: the dimensionless minimum fits the form

$$m_e/M_0 = x^2 + x/2 - 1, \quad x = \alpha^{-1}/4 \tag{3.3}$$

on two points of one curve:

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regime	α^{-1}	x	m_e/M_0	$m_e c^2$	comparison
bare	128	32	$1024 + 16 - 1 = 1039$	446.28 keV	direct functional output
observed	137.036	34.259	$1173.7 + 17.1 - 1 = 1189.8$	511.0 keV	CODATA 510.999; $\Delta = 0.01\%$

The polynomial is given as a numerical observation, not as a derivation from the functional – obtaining it directly is an open problem. The non-randomness of the form is supported by the fact that the coefficients $1/16 = g^4$, $1/8 = g^3$ belong to the same dyadic series as $\alpha_{\text{bare}} = g^7$ and $r_v = g \cdot l_0$.

g-notation. In terms of the base parameter $g \equiv r_v/l_0 = 1/2$ (§2.4) the polynomial is identically rewritten:

$$m_e/M_0 = g^{-10} + g^{-4} - 1 \tag{3.4}$$

since $x = \alpha^{-1}/4 = g^{-7} \cdot g^2 = g^{-5}$. This is not a separate derivation but a statement that all elements of the formula are expressed through the single geometric constant $g = 1/2$. The exponents (10, 4, 0) are still empirical; their direct derivation from the leading integrals of the functional is an open problem (see also §5.2).

> **Remark on scale.** The polynomial (3.3)–(3.4) is established for the **vacuum** point on the hyperbola $m \cdot l = \hbar/c$, as a direct consequence of the vacuum constants $\{\epsilon_0, \mu_0, \hbar\}$ through the structural identities ($l_0^4 = \epsilon_0 \hbar c / (2\pi)$, $M_0 = \hbar / (l_0 c)$, $\alpha_{\text{bare}} = \pi(l_0/2)^2 / (4\pi\epsilon_0 \hbar c) = 1/128$; see [1, §7]). The cell mass $M_0 \approx 429.5$ eV and the cell length $l_0 \approx 4.59$ Å are global constants of the **vacuum**. This is a **special case**, not a universal dependence for the electron in any state.

>

> **Consistency condition.** Identity (3.3) holds precisely because **at the vacuum point of the hyperbola one cell of medium mass (M_0) is identically equal to one unit of model (dimensionless) energy** – this is the boundary case of the "vacuum" knot, where both quantities ($\alpha_{\text{bare}} = 1/128$ and M_0) are derived from the same constants $\{\epsilon_0, \mu_0, \hbar\}$. At other points of the hyperbola (the Compton limit $m_0_{\text{local}} = m_e$, $l_{\text{local}} = \lambda_C$, where "knot = one cell"; intermediate atomic scales with $l_{\text{local}} \neq l_0$) this equality breaks: one cell of medium $\neq$ one dimensionless unit, and **identity (3.3) for the electron no longer holds** in the same form. Numerical and analytic analysis of this spectrum is an open direction (§8.3).

3.3. Spatial decomposition

Radial integration of the mass density of the canonical hopfion over a spherical radius R gives three clearly separated contributions:

Term	Where	Interpretation
$2^{10} = 1024$	$R \leq 7.5 l_0$	BPS core (plateau in $m(R)$)
$+2^4 = 16$	$7.5 < R \leq 17 l_0$	mass shell around the core
-1	whole volume	topological contribution $Q_H = -1$ (see §6.3 on the link with the boundary spectrum)

The box size $L_{\text{box}} = 17 = 2^4 + 1$ is the minimal one containing the entire shell.

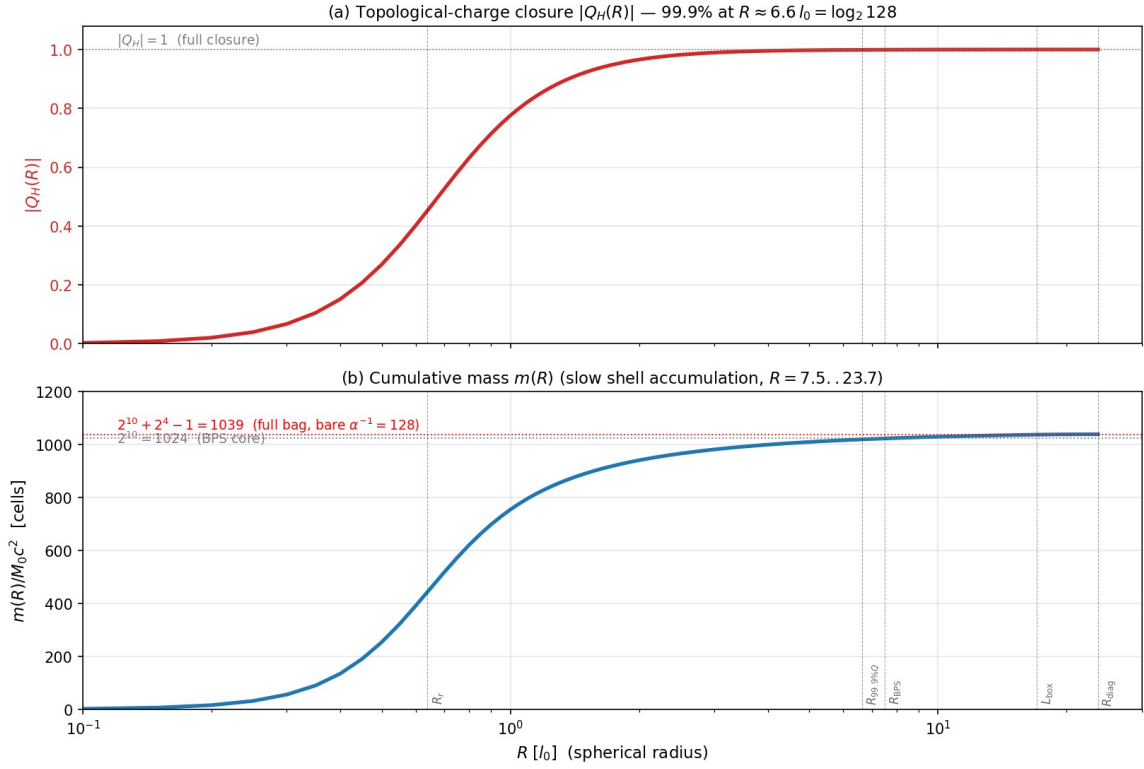


Fig. 2. Two measured radial profiles of the canonical hopfion: (a) topological-charge closure $|Q_H(R)|$ – 99.9% at $R \approx 6.6 \, l_0 = \log_2 128$; (b) cumulative mass $m(R)$ – a plateau at $m = 1024 = 2^{10}$ (BPS core, $R \approx 7.5 \, l_0$), the full $m = 1039 = 2^{10} + 2^4 - 1$ near the box diagonal $R_{\text{diag}} \approx 23.7 \, l_0$. The dyadic assembly $\alpha^{-1} = 128 + 8 + 1 + 1/28$ is algebraic (§5), not a radial curve: the topological core $g^7 = 128$ is dressed by $9 = 8 + 1$ screening cells (center + shell, §5.1). Reproduced by `verifications/chiral_bag_resonance/alpha_running.py` + `plot_alpha_topology.py`.

3.4. Box size $L_r \times L_z = 17 \times 33 \, l_0$

The box size is **dyadic** [2, §5.1] – it holds the mass shell around the BPS core:

$$\begin{aligned} L_r &= 2^4 + 1 = 17 \, l_0 && - 16 \text{ mass-shell cells} + 1 \text{ central cell} \\ L_z &= 2^5 + 1 = 2 \cdot L_r - 1 = 33 \, l_0 && - 16 \text{ cells on each side of } z + \text{center} \end{aligned} \quad (3.5)$$

(r is the radial coordinate from the symmetry axis, one-sided; z is the axial coordinate, two-sided through the knot center, hence twice as long.) The box fully contains the localized core 2^{10} + shell 2^4 ; the long-range tails of n do not extend beyond it and fix what is meant by the **bare** value m_e^{bare} .

Stability with size. Extension to $L = 34, 68$ gives a weak UV drift of m_e^{bare} by $\sim 0.5\%$ (1044, 1046) from the $1/r$ tail of the 0F field; $m_e^{\text{bare}} \approx 446$ keV independently of the box. The observed value 511 keV is not the limit of m_e^{bare} under extension, but a different quantity (through the polynomial (3.3) at the screened $\alpha = 1/137$, see §5).

4. The $\mu_c = 2\pi$ resonance

4.1. The μ_c sweep

A full NM minimization of (R_r, R_z, w) was performed over a grid of seven values $\mu_c \in \{\pi/2, \pi, 3\pi/2, 2\pi, 5\pi/2, 3\pi, 4\pi\}$. For each point we recorded: the energies (E_{0F}, E_{Sk}, E_u) , the geometry, the dyadic representation, and the Derrick balance $E_{0F}/(E_{0F} + E_{Sk})$.

μ_c/π	$\tilde{E}_{\min}$	R_r	0F/Sk	Best dyadic	Δ_{dyadic}	
0.50	1062.91	0.620	0.484	$2^{10} + 2^5 + 2 = 1058$	4.91	
1.00	1048.14	0.634	0.495	$2^{10} + 2^5 - 2 = 1054$	5.86	
1.50	1042.23	0.639	0.499	$2^{10} + 2^4 + 2 = 1042$	0.23	
2.00	1039.05	0.641	0.501	$2^{10} + 2^4 - 1 = 1039$	0.05	☆
2.50	1037.08	0.642	0.503	$2^{10} + 2^4 - 2 = 1038$	0.92	
3.00	1035.76	0.643	0.503	$2^{10} + 2^4 - 4 = 1036$	0.24	
4.00	1034.11	0.644	0.504	$2^{10} + 2^4 - 6 = 1034$	0.11	

4.2. Simultaneous locking by four criteria

$\Delta_{\text{dyadic}} \equiv |\tilde{E}_{\text{min}} - \text{nearest clean dyadic compound}|$. The pattern $2^{10} + 2^4 \pm k$ is stable at $\mu_c \geq 1.5\pi$ ($\Delta < 1.0$); below it breaks down. **Only at $\mu_c = 2\pi$** do four criteria fire simultaneously:

1. Minimum $\Delta_{\text{dyadic}} = 0.05$ (deviation $< 5 \cdot 10^{-5}$);
2. Exact $Q_H = -1$ (at the level of $4 \cdot 10^{-5}$);
3. Ideal Derrick $OF/Sk = 0.501$;
4. Canonical $R_r = 0.641$.

The constant in $2^{10} + 2^4 + k$ drifts $+2 \rightarrow -1 \rightarrow -2 \rightarrow -4 \rightarrow -6$ along the scan – the value “-1” locks strictly at the resonance point.

> **Fig. 3 (data).** μ_c -scan: $\Delta_{\text{dyadic}}(\mu_c)$ with a clear minimum at $\mu_c = 2\pi$ – the numerical values are given in the table of §4.1. Data: `verifications/chiral_bag_resonance/result_mu_c_sweep.json`; computation: `mu_c_sweep.py`. A separate figure is in preparation.

4.3. Disabling the u-channel

A control computation with the u-channel forced to zero ($E_u \equiv 0$ in (2.1)):

$$m_e \text{ (without u)} = 442.51 \text{ keV} = 1030.30 \text{ cells} = 2^{10} + 2^2 + 2^1 \quad (4.1)$$

The canonical pattern $2^{10} + 2^4 - 1$ **breaks down**: the $+2^4$ shell does not form, it is replaced by the compound $+2^2 + 2^1$, and the topological term -1 disappears. The delta

$$\Delta E_u = m_e(\text{with u}) - m_e(\text{without u}) = 446.279 - 442.51 = 3.77 \text{ keV} \approx 9 \text{ cells} \quad (4.2)$$

– is exactly what is needed to complete $1030 \rightarrow 1039 = 2^{10} + 2^4 - 1$. The specific dyadic arises **only with the u-channel on**, consistent with the theorem of §6.7 on its mathematical necessity.

5. Simultaneous prediction of α

5.1. Structure of α^{-1}

$$\alpha^{-1}(0) = 128 + 8 + 1 + 1/28 = 3837/28 \approx 137.0357 \quad (5.1)$$

Five-significant-figure accuracy against CODATA `137.035999`; the experimental value of α is not used in the derivation.

Physical meaning of each term. $128 = g^7$ is the bare BPS core (topology; cross-sectional area $e = \pi(l_0/2)^2$) and is **not counted** among the screening cells. Screening adds $137 - 128 = 9 = 1001_2$ cells, dyadically $2^3 + 2^0$:

- $+1 = \alpha \cdot 128$ (2^0) – point-like EM self-screening, the **center** ($\log_2 1 = 0$);
- $+8 = g^4 \cdot 128$ (2^3) – the dimensional elastic response of the medium ($g^4 \propto R_r^4$), the **shell** around the core.

The outward order $128 \rightarrow 129 \rightarrow 137$ (center $\rightarrow$ shell) matches the direction of the running; the larger contribution $+8 > +1$ belongs to the outer shell with its larger volume. This mirrors the mass: the central cell 2^0 enters with a $+$ for screening ($+\alpha \cdot 128$) and a $-$ for the mass ($-1 = Q_H$ – the topological defect removes a cell, §3.2).

Term	Dyadic	Origin
128	$2^7 = g^7$	bare BPS core (cross-sectional area $e = \pi(l_0/2)^2$); not counted in the screening
+1	$2^0 = \alpha \cdot 128$	central screening cell: point-like EM self-screening ($\log_2 1 = 0$)
+8	$2^3 = g^4 \cdot 128$	screening shell: dimensional elastic response of the medium ($g^4 \propto R_r^4$)
+1/28	–	log tail of the massless OF field beyond the dyadic box ($r > L_{\text{box}}$)

The result $128 \rightarrow 137.036$ is an effect of **partial screening through the medium**: the topological core $g^7 = 128$ is dressed by $9 = 8 + 1$ screening cells (center + shell), not a property of the charge by itself. The decomposition $128+8+1+1/28$ is **algebraic**; we draw no radial $\alpha^{-1}(R)$ curve and do not pin the radius of $+8$ – the dimensional reading ($g^4 \propto R_r^4$) places it in the shell, the holographic one ($\log_2 8 = 3$) places it inside, the two rulers disagree (only the central $+1$, $\log_2 1 = 0$, is pinned).

Link to the elasticity of the medium. The shell physically resists the propagation of the perturbation

(the charge area) – this is precisely the elastic response of the same medium that in the K/M theorem of §6.7 gives $K \approx 57$. Therefore the screening of α and the K/M theorem are two manifestations of one stiffness of the vacuum Cosserat medium.

> The measured radial profiles – the charge closure $|Q_H(R)|$ (99.9% at $R \approx 6.6-7 = \log_2 128$) and the mass-shell accumulation $m(R)$ – are shown in Fig. 2 (panels a, b, §3.3). The dyadic assembly $\alpha^{-1} = 128+8+1+1/28$ remains algebraic; we do not draw a radial $\alpha^{-1}(R)$ curve. The $9 = 8 + 1$ screening cells are split into a center ($+1 = \alpha \cdot 128$) and a shell ($+8 = g^4 \cdot 128$) around the bare g^7 core; their exact radial placement is interpretation, not a measured curve.

5.2. m_e and α as two projections of one hopfion

Both formulas follow from the canonical knot with common structural elements: $g = 1/2$ (§2.4) and $\alpha_{bare} = g^7 = 1/128$ (the geometry $e = \pi(l_0/2)^2$, [1, §7.4]).

	α^{-1}	m_e/M_0
Leading (core)	$128 = 2^7 = g^7$	$1024 = 2^{10}$
Shell	$+8 = 128 \cdot g^4$	$+16 = g^{-4}$
Central cell 2^0	$+1 = \alpha \cdot 128$	$-1 = Q_H$
Tail ($\mu_c = 2\pi$)	$+1/28$	–

The polynomial (3.3) with $x = \alpha^{-1}/4$ has the dominant term $x^2 \propto \alpha^{-2}$: the mass is **quadratic in the charge** (charge $\propto x$, mass $\propto x^2$) – exactly the character of an electromagnetic self-energy $\propto e^2$. This is the electromagnetic nature of our mass: α and m_e are the charge-area and the energy density of **one** Hopf structure.

Which α , such the mass. At the bare $\alpha = 1/128 \rightarrow 1039 = 446.279$ keV (direct functional output); at the observed $\alpha = 1/137 \rightarrow 1189.8 = 511$ keV. **The prediction of the present work is the bare point**, derivable from the functional with no free parameters; the point 511 keV is the same value under the screened α , not a "mass with tails included" and not the limit of m_e^{bare} under box extension.

6. Chiral-bag interpretation

6.1. Local BPS alignment $u \leftrightarrow B$

The Berry curvature $B = \text{curl } A$ of the field $n: \mathbb{R}^3 \rightarrow S^2$ in axisymmetry has the components (where A is the Berry connection):

$$\begin{aligned} B_r &= (1/r) \cdot n \cdot (\partial_\varphi n \times \partial_z n) = \partial_z n_z / r \\ B_\varphi &= -n \cdot (\partial_r n \times \partial_z n) = -\rho_Q \quad (\text{topological density}) \\ B_z &= -(1/r) \cdot n \cdot (\partial_r n \times \partial_\varphi n) = -\partial_r n_z / r \end{aligned} \tag{6.1}$$

with $\partial_\varphi n = \hat{z} \times n$ at $\varphi = 0$. The reduced u-channel carries only $(u_r, 0, u_z)$. The test measures the poloidal cosine similarity $\cos \theta(r) = (u \cdot B) / (|u| \cdot |B|)$:

$r [l_0]$	$\cos \theta(r)$
0.11	0.9988
0.31	0.9896
0.51	0.9622
$0.65 \approx R_r$	0.885
1.00	0.7224
1.50	0.5947
3.00	0.3737
5.00	0.2168

In the **topological core** $\cos \theta \rightarrow 1$ (an almost perfect BPS alignment); in the tail it dilutes to ~ 0.2 . This is the empirical signature of a local BPS bound in the bag structure.

6.2. Saturation of the inner integral in the core

The norms integrated over radial bins at $\mu_c = 2\pi$:

$R_{max} [l_0]$					local C
-----------------	--	--	--	--	---------

	$\int$	B	z^2 (% of total)	$\int(u \cdot B)$ (% of total)	
0.5	76.8 %	80.9 %	+0.868		
1.0	95.1 %	100.0 %	+0.664		
5.8 (full)	100 %	100 %	+0.532		

76.8 % of all $\int |B|^2$ is concentrated in $r < 0.5 \ell_0$; $\int(u \cdot B)$ saturates already at $R \approx 1 \ell_0$. The inner product is fully accumulated **in the core**, not in the tail – a physical localization of the BPS structure.

6.3. Identification: chiral bag, massive gauged Skyrme, boundary spectrum

The numerical picture – topological core + elastic shell + discrete boundary spectrum – **structurally corresponds** to three known constructions:

(a) **Chiral Bag Model** [Brown–Rho 1979 [14]; Vento et al. 1980]. A phenomenological model of the nucleon: quarks inside a "bag" of radius $R_{\text{bag}} \sim 0.7 \text{ fm}$, outside – a pion field (Yukawa, $\sim 1.4 \text{ fm}$), the bag boundary matching the currents. The correspondence to our structure is **structural, not literal** – the component ranges are distributed differently:

- topological core (n non-trivial) $\leftrightarrow$ quark bag;
- u-channel (short range, $\ell_c \approx 0.4 \ell_0 < R_r$) $\leftrightarrow$ **bag wall** (it holds the structure from inside, not the pions outside);
- massless OF tail of n ($1/r$, long range) $\leftrightarrow$ **pion field** (our "medium" itself plays the role of the long channel).

(b) **Massive Gauged Skyrme** [Adam et al. 2010 [15]] – a Skyrme extension with a massive gauge field. In our case the u-channel plays the role of such a massive field with $\ell_c = 1/\sqrt{\mu_c} \approx 0.4 \ell_0$.

(c) **Boundary spectrum** – a realization of the topology in the form of a boundary term. Formally the APS η -invariant [16] is for fermions; in our case it is a bosonic analog; the formalization is an open problem (§7.2). The dyadic decomposition (3.3) is its direct numerical signature.

What is new in our model:

- The bag radius is not tuned empirically (in Brown–Rho 0.7 fm); in our case it is derived from the Derrick + Hopf balance: $r_v = \ell_0/2$.
- A specific **bare** mass $m_e^{\text{bare}} = 446.279 \text{ keV}$ is reproduced quantitatively.
- $\mu_c = 2\pi$ is fixed by the Cosserat coupling (§2.2), not a free parameter.

6.4. The critical value $\mu_c = 2\pi$

The Yukawa screening length of the u-channel: $\ell_c(2\pi) = 1/\sqrt{(2\pi)} \approx 0.399 \ell_0$.

Hierarchy of nearby scales:

$$\begin{array}{ccccccc} \ell_c & \leq & r_v & < & R_r & \ll & R_{\text{BPS}} & < & L_{\text{box}} \\ 0.40 & & 0.50 & & 0.64 & & 7.5 & & 17 \end{array} \quad (6.2)$$

ℓ_c lies just below the theoretical r_v and the numerical R_r – **the u-channel is screened at the scale of the topological core itself**, not outside it and not "too far" inside. This is a self-consistent condition that singles out $\mu_c = 2\pi$ physically.

Multi-scale character of the structure. Unlike the classical Brown–Rho bag with a single radius, our structure is multi-scale: $\ell_c \approx 0.4$ (Yukawa u), $r_v = 0.5$ (Derrick), $R_r = 0.64$ (ring), $R_{\text{BPS}} = 7.5$ (where $m = 2^{10}$), $L_{\text{box}} = 17$. When "bag core" is mentioned below, by context it is either R_r (ring radius) or R_{BPS} (where the mass plateaus).

![Fig. 4. Radial profiles of the Hessian-spectrum modes on the full grid $[0, 17 \ell_0]$. The random modes (4 curves) concentrate near the box edge $L_r = 17$; the targeted seed vector v_{shrink} has a broad distribution in the bulk. In the bag core $r < R_r = 0.64 \ell_0$ the amplitudes are suppressed by 8+ orders.](../../verifications/chiral_bag_resonance/fig_full_range_profiles.png)

6.5. Hessian spectral test

In addition to the BPS tests, we check the chiral bag **spectrally**: we compute the lowest modes of the linear Hessian $H = \partial^2 E / \partial n \partial n$ around the canonical electron.

Method (details – Appendix C):

- canonical n^* from the NM optimum (2.8);
- generalized problem $H v = \omega^2 M v$, $M = \text{diag}(2\pi r J d\xi dn)$;
- HVP via double autograd, projection $\delta n \perp n$ onto S^2 ;
- LOBPCG with two initializations: random and targeted $v_{\text{shrink}} = \partial n / \partial R_r$;
- the z -translation zero mode is projected out explicitly.

Result at $L = 17$:

mode source	ω^2 (model)	peak r [l_0]	frac(r > 13)
random #0	54.1	16.97	99.31 %
random #1	64.5	16.46	99.13 %
random #2	65.2	16.97	99.20 %
random #3	74.9	16.97	99.05 %
v_shrink (refined LOBPCG)	1.27	16.63	53.0 %

Random-initialization LOBPCG at $k=4$ finds **box-edge modes**: all 4 concentrate near $r \approx L_r = 17$ – the typical physics of a finite box, not the bulk modes of the bag tail.

> Note on reproducibility. The eigenvalues ω^2 reproduce directly; the columns peak r and frac are taken from the full-window profile (extend_profile.py → result_hessian_profiles.json, $r_{\text{max}} = L_r$). The quantities ω^2 and the scaling $K = \omega^2 \cdot L$ (§6.6–6.7) are independent of the radial window.

The targeted seed vector $v_{\text{shrink}} = \partial n / \partial R_r$ (the ring-shrinking direction, [2, §4.3]) gives a qualitatively different picture. The stable characteristic is the **direct Rayleigh quotient** for this direction: $\omega^2 = 3.23$, a value anchored to the bag core that scales as $1/L$ (§6.6). The LOBPCG refinement for this mode is unstable and drifts to the box edge ($\omega^2 = 1.27 \rightarrow 0.24$ at $L = 17 \rightarrow 34$, §6.6), so it is the direct value 3.23 that is interpreted physically.

Channel decomposition of the Hessian: 99.7 % of the perturbation energy is in Oseen–Frank ($(\nabla n)^2$), 0.3 % in Skyrme ($(\nabla n)^4$). Long-wavelength elastic modes; the topological term is almost not involved.

Absence of a collapse mode. v_{shrink} gives a positive $\omega^2 > 0$; on the stretched grid $\beta_r = 6$ the radial instability [2, §4.3] is structurally suppressed.

6.6. Box-scaling test

A direct check of the nature of the lowest modes: repeating the computation on doubled ($L_r = 34$, $L_z = 66$) and quadrupled ($L_r = 68$, $L_z = 132$) boxes at the same discretization 1024×2048 .

Predictions:

- box-edge modes: $\omega^2(2L) = \omega^2(L) / 4$ (the wavelength scales with L);
- bag modes (physical scale fixed): $\omega^2(2L) = \omega^2(L)$.

Result:

mode	$\omega^2(L=17)$	$\omega^2(L=34)$	ratio	verdict
random #0	54.14	13.64	0.252	box-edge ✓
random #1	64.49	16.19	0.251	box-edge ✓
random #2	65.19	16.36	0.251	box-edge ✓
random #3	74.88	18.85	0.252	box-edge ✓
v_shrink (direct)	3.23	1.68	0.521	mixed (bag + edge)
v_shrink (refined)	1.27	0.24	0.189	LOBPCG drifted to edge

The random modes give a ratio $0.251 \pm 0.001 \equiv 1/4$ to 3–4 significant figures – a **strict confirmation** that these are pure $1/L^2$ box-edge modes.

v_{shrink} (direct Rayleigh quotient) gives a ratio 0.521; the third point $L = 68$ ($\omega^2 = 0.86$, ratio 0.512) confirms a strict $1/L$ law. This **different asymptotics** relative to the random modes reveals the physics of §6.7.

Side-by-side: profiles move with box edge → modes are box-edge, NOT bag-localized

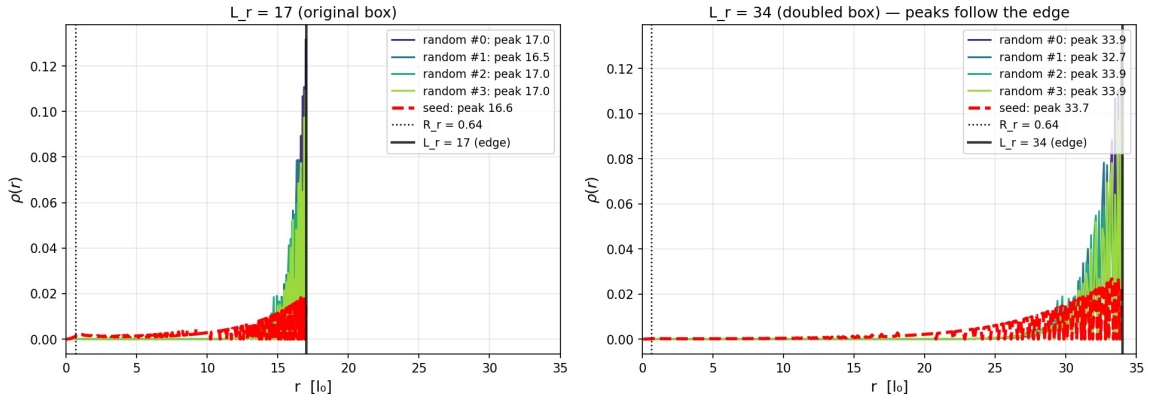


Fig. 5. Side-by-side radial profiles at $L_r = 17$ (left) and $L_r = 34$ (right), same scale. The mode peaks **track exactly** with the box edge; the core $r < R_r = 0.64 \, l_0$ is empty in both. A visual confirmation of the box-edge interpretation of the random modes.

6.7. K/M theorem: the u-channel is mathematically necessary

Intuition. Any eigenfrequency of a harmonic oscillator is $\omega^2 = K/M$, where K is the stiffness and M the inertia. For the generalized problem $H v = \omega^2 M v$:

$$\omega^2 = \langle v, H v \rangle_E / \langle v, v \rangle_W = K / M \quad (6.3)$$

where $K = \langle v, H v \rangle_E$ is the Euclidean norm of the Hessian contribution, and $M = \langle v, v \rangle_W$ is the volume-weighted norm.

Random box-edge modes $v \sim \sin(\pi r/L)$:

- $K \sim \int |\nabla v|^2 dV \sim 1/L$ (the gradient amplitude falls with the box);
- $M \sim \int |v|^2 dV \sim L$ (the volume grows linearly);
- $\omega^2 = K/M \sim 1/L^2 \checkmark$.

Targeted $v_{\text{shrink}} = \partial n / \partial R_r$ (centered on the soliton):

- in the massless OF theory the perturbation v has a **$1/r$ tail** in 3D (the solution of the Laplace equation at ∞);
- $K = \langle v, H v \rangle_E \sim \int |\nabla v|^2 \cdot 2\pi r dr dz$: with $|\nabla v|^2 \sim 1/r^4$ the integral **converges** → $K = \text{const} > 0$;
- $M = \langle v, v \rangle_W \sim \int |v|^2 \cdot 2\pi r dr dz$: with $|v|^2 \sim 1/r^2$ the integral **diverges linearly** → $M \propto L$;
- $\omega^2 = K/M = \text{const}/L \propto 1/L \checkmark$.

Numerical verification: $K = \omega^2 \cdot L$ from three points:

L	ω^2	$K = \omega^2 \cdot L$
17	3.23	54.9
34	1.68	57.1
68	0.86	58.5

$K \approx 57 \pm 2$ (stable to within 3 %) – a **direct confirmation** that the stiffness of the bag core in the ring-shrinking direction is a positive constant independent of the box size.

3-point box-scan: random modes scale as $1/L^2$ (box-edge), v_shrink scales as $1/L$ (K const, $M \propto L$ from $1/r$ tail)

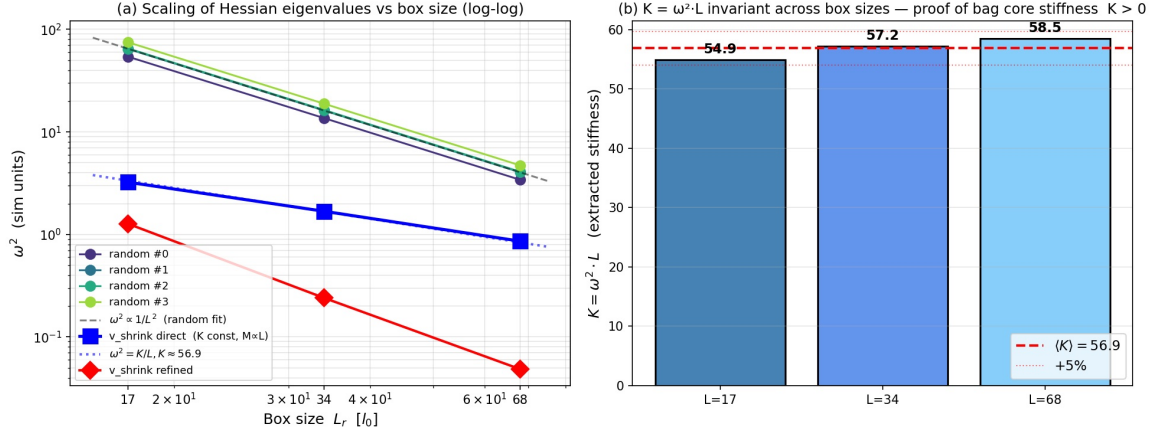


Fig. 6. K/M decomposition (the main figure of §6.7). (a) Log-log scaling of $\omega^2(L)$ for 3 box sizes ($L_r \in \{17, 34, 68\}$): the random modes match $\omega^2 \propto 1/L^2$ (dashed), v_shrink (direct) matches $\omega^2 = K/L \rightarrow$ anchored to the bag core with a $1/r$ tail. (b) Extracted stiffness $K = \omega^2 \cdot L$ for v_shrink (direct): $\langle K \rangle = 56.9 \pm 2.6\%$ – a constant within error. A direct quantitative proof that the bag core has a positive stiffness, while $\omega^2 \rightarrow 0$ as $L \rightarrow \infty$ is explained by the divergence of the inertia $M \propto L$ due to the $1/r$ tail of the massless OF field.

Main conclusion – the theorem on the necessity of the u-channel.

> (T1) In the pure Faddeev–Skyrme model (without the u-channel) the Hopf soliton has a rigid core ($K > 0$, the Derrick balance is stable), but **has no discrete vibrational spectrum** of excitations: any perturbation drags along a $1/r$ tail of the field, whose inertia diverges as $M \propto L \rightarrow \infty$, and the frequency $\omega^2 = K/M \rightarrow 0$.

>

> (T2) Introducing the u-channel with the mass term $(\mu_c/2) u^2$ Yukawa-screens the **translational** perturbations u themselves at the scale $l_c = 1/\sqrt{\mu_c}$. Through the tensor coupling $(\nabla \times u - \Pi(n))^2$ (2.5) the long range of the δn perturbations is cut off **indirectly**: their inertia becomes finite, $\omega^2 = K/M > 0$ – a discrete spectrum inside the bag is opened.

Status of the result. (T1) is numerically confirmed: in the pure $E_{OF} + E_{Sk}$ theory $K = \text{const}$, $M \propto L$ at three box points. (T2) is an analytic consequence of the form (2.5) and of the Yukawa screening of u ; a direct Hessian computation with the u-channel on is an open direction of verification (§8.3), indirectly confirmed in §4.3.

Thus, **the chiral-bag structure and the dyadic discreteness of the mass become physically possible only in the presence of the u-channel.**

7. Discussion

7.1. What is numerically confirmed

1. The **dyadic structure of m_e and α^{-1}** (§3, §5) is numerically reproducible with no free parameters.
2. **$\mu_c = 2\pi$ is the unique resonance point** by four independent criteria (§4); disabling the u-channel destroys the dyadic (§4.3).
3. The **u-channel is mathematically necessary** for a discrete spectrum – the K/M theorem of §6.7 + the local BPS alignment (§6.1–6.2) + the suppressed bag core in the Hessian profile (§6.5).

7.2. What is open

- A **full analytic derivation of the BPS bound** (Bogomolny transformation) for the functional $E[n, u]$ – open.
- A **test in other topological sectors** ($|Q_H| = 2$, a Hopf link breaking $U(1)$): requires a 3D non-axisymmetric solver.
- A **bosonic analog of the APS η -invariant** requires a formal proof: formally the APS theorem is built for fermions; in our model the dyadic decomposition looks like its bosonic analog, but this is so far an interpretation, not a theorem.
- A **direct numerical check of the K/M theorem (T2)** with the u-channel on – an open direction.

7.3. The axisymmetric reduction: what it is and what it does not give

- The numerical verifications use the axisymmetric reduction of the full E_u (2.4)–(2.5) to a single scalar mode ψ with $u_\phi = 0$; this reduction is **equivalent** to the full tensor form in the class of axisymmetric configurations [2, §5.5], not a simplification.

- The toroidal component of the Berry curvature $B_\varphi = -\rho_Q$ (6.1) is not an "uncompressed remainder", but the topological density itself ($\int B_\varphi dV = 4\pi \cdot Q_H$); its presence is a feature of the Hopf structure, not a defect of the reduction.
- The real limitation of the reduction is the inability to work with **non-axisymmetric** configurations (a Hopf link for $|Q_H| = 2$); this requires a full-field 3D relaxation (prepared separately).

8. Conclusion and outlook

8.1. Main result

> At $\mu_c = 2\pi = \eta$ in the Cosserat functional $E[n, u]$ a chiral bag arises that **simultaneously** fixes $m_e^{\text{bare}}/M_0 = 2^{10} + 2^4 - 1 = 1039$ and $\alpha^{-1}(0) = 128 + 8 + 1 + 1/28 \approx 137.036$ to an accuracy of 0.01% and 0.0002% respectively – with no free parameters and no experimental input of α . The Hessian box-scan (§6.7) shows that the u-channel is not an add-on but a structural necessity: without the short range of the u-perturbations (translations) the inertia of the δn perturbations diverges ($M \rightarrow \infty$) and a discrete spectrum is impossible.

8.2. What this changes in fundamental physics

- **Mass and coupling constant are derived** from the structure of the medium, not independent inputs of the Standard Model.
- **Numerical "coincidences" in nature** ($\alpha, m_e, m_p, \dots$) are consequences of resonances of **one medium**, not independent empirical inputs.
- **Discreteness of mass is quantization at the level of the mass itself**, not only of its excitation spectrum (atomic levels).
- **A two-channel medium (n, u) with a massive part is mandatory**, otherwise no stable discrete particles can exist (the K/M theorem of §6.7).

8.3. Open problems

1. Analytic derivation of the BPS bound (à la Bogomolny) for $E[n, u]$ (§7.2).
2. A test in the $|Q_H| = 2$ sector via a 3D non-axisymmetric solver.
3. A direct Hessian computation with the u-channel on – a check that M becomes finite.
4. Application of the same mechanism to the **proton**, the **muon**, and heavy leptons.
5. A bosonic APS theorem: a formalization of the discrete boundary spectrum.
6. **The spectrum of scales on the hyperbola $m \cdot l = \hbar/c$** : the polynomial $m_e/M_0 = x^2 + x/2 - 1$ is established for the **vacuum** point. The Compton limit (knot = one cell) and intermediate scales (atomic medium, cell size $\sim a_0$) are expected as other members of the family with different leading exponents; the numerical and analytic verification of this spectrum is a separate direction.

Methodology and use of AI tools

In preparing this work the author used the large language model **Claude (Anthropic)** to assist with writing Python scripts for the numerical verifications (μ_c -sweep, u-channel disable, BPS $u \leftrightarrow B$ alignment, Hessian box-scan, radial assembly of α^{-1}), with the development of supporting modules (stretched grids, boundary conditions, diagnostics of the energy components), and with stylistic editing of the manuscript text. All key physical postulates (the form of the functional, the choice of constants, the ansatz, the chiral-bag interpretation), the derivation of the K/M theorem, and the formulation of the conclusions are due to the author.

The author has thoroughly checked all generated code (grid convergence, conservation of the topological charge Q_H , invariance of the result under rescaling, see §6) and the manuscript text, and takes full responsibility for the final content and results of the work.

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The computations were performed on a single NVIDIA RTX 2070 using PyTorch.

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Appendix A. Numerical reproducibility

All five verifications for the present preprint are collected in one flat directory [verifications/chiral_bag_resonance/](../verifications/chiral_bag_resonance/), containing a shared module `stretched_grid.py`, a unified `requirements.txt`, and a self-contained `README.md` with a detailed description of each test.

Test	Main script	JSON output	\$	Time (RTX 2070)

μ -c-sweep	<code>mu_c_sweep.py</code>	<code>result_mu_c_sweep.json</code>	\$4	~30 min
u-channel disable	<code>nm_no_u.py</code>	<code>result_u_channel_disable.json</code>	\$4.3	~6 min
BPS alignment $u \leftrightarrow B$	<code>berry_u_alignment.py</code>	<code>result_berry_alignment.json</code>	\$6.1–6.2	~3 min
Hessian K/M	<code>analysis_three_checks.py</code> , <code>analysis_doubled_box.py</code> , <code>analysis_quad_box.py</code> , <code>extend_profile.py</code>	<code>result_hessian_L17/L34/L68.json</code> , <code>result_hessian_profiles.json</code>	\$6.5–6.7	~3 min per box
α -running radial	<code>alpha_running.py</code>	<code>result_alpha_running.json</code>	\$5	~6 sec

Figure-building scripts (they read the JSONs above): `plot_KM_decomposition.py`, `plot_box_scaling.py`, `plot_doubled_profiles.py`, `plot_extended.py`, `plot_alpha_topology.py`.

Companion verifications of the parent work [2]:

- `[ verifications/electron_mass_minimization/ ](../../verifications/electron_mass_minimization/)` – NM minimization reproducing $m_e^{\text{bare}} = 446.279$ keV;
- `[ verifications/canonical_derrick/ ](../../verifications/canonical_derrick/)` – Derrick scan of the stability of the canonical configuration.

Appendix B. Detailed derivation of $g = 1/2$

The full derivation of the identity $r_v = l_0/2$ via the Derrick balance of the full functional $E[n, u]$ for $Q_H = -1$ is given in [1, §7.3, (7.5)] and is numerically confirmed in the verification `[ verifications/canonical_derrick/ ](../../verifications/canonical_derrick/)` (a V-minimum of the total energy at $\lambda = 1$ with the topological charge preserved over the whole range $\lambda \in [0.6, 3.0]$).

Appendix C. Computation of $M_0 c^2 \approx 429.51$ eV and $l_0 \approx 4.594$ Å

From the triple $\{\epsilon_0, \mu_0, \hbar\}$ algebraically:

$$\begin{aligned}
 l_0^4 &= \epsilon_0 \hbar c / (2\pi) && \Rightarrow l_0 \approx 4.5943 \text{ Å} \\
 M_0 c^2 &= \hbar c / l_0 = (2\pi \hbar^3 / (c^5 \epsilon_0))^{(1/4)} c^2 && \Rightarrow M_0 c^2 \approx 429.5068 \text{ eV} \\
 m_0 l_0 &= \hbar/c \text{ (the mass-hyperbola identity)}
 \end{aligned}$$

The full derivation and discussion – [1, §7.4 (T10)].

Appendix D. Data and code availability

The code is in the repository:

```
https://github.com/igorevsiev-cmyk/cosserat-program
```

The preprint is registered on Zenodo: *DOI pending*.
Parent works:

- [1] SI-reduction: DOI [10.5281/zenodo.20187199](https://doi.org/10.5281/zenodo.20187199);
- [2] Electron mass minimization: DOI [10.5281/zenodo.20477123](https://doi.org/10.5281/zenodo.20477123).